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AI-Driven Real-Time Intrusion Detection and Automated Mitigation for Industrial IoT Process Control Sensors

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***Abstract*— This research proposes an Optimized XGBoost-based Intrusion Detection System, O-XGB IDS, to provide improved accuracy, reduce latency, and guarantee scalability in diverse IoT environments. The developed model integrates Bayesian optimization of hyperparameters along with adaptive feature selection to improve the classification performance. For this, experimental validation is conducted on multiple benchmark datasets like IoT-23, RT-IoT2022, and MQTT-IoT-IDS2020 by comparing different machine learning models such as Decision Tree, Random Forest, SVM, and Logistic Regression. These results demonstrate that the proposed O-XGB achieved the highest accuracy in intrusion detection of 98.93%, precision of 98.40%, and a 29% faster response time compared with other ensemble-based models. It has been confirmed that the optimized XGBoost framework furnishes a robust, low-latency, highly reliable defense mechanism for real-time IoT cybersecurity applications, opening up new research dimensions toward intelligent and adaptive intrusion detection systems.**

Keywords— **IoT Security, Process Control Sensors, Cyber Attack Detection, XGBoost, Machine Learning, False Data Injection, MQTT, CoAP, Real-Time Response, Industrial IoT**

I. Introduction

Industrial automation has changed with the growth in the connectedness of devices, forming the Internet of Things intelligent spaces with real-time monitoring and adaptive control. In such a new paradigm, process control sensors play a major role for maintaining efficiency and safety in an industrial system. Commonly, such devices use lightweight communication protocols like MQTT and CoAP that enable efficient and low-latency data exchange among the nodes of a distributed network. However, their inability to handle complex computations and lack of appropriate security mechanisms make them highly susceptible to various cyber-attacks such as FDI, manipulation of commands, and DoS attacks [1].

In this perspective, among the AI-based techniques, Extreme Gradient Boosting has been regarded as a high-performance ensemble technique; hence, it has resulted in efficient processes in handling complex, nonlinear, and high-dimensional IoT data [4]. However, the standard implementations of XGBoost have often presented the problem of being sensitive to hyperparameters, computational overhead, and difficult adaptation in a streaming data environment [5]. In this regard, the optimization of XGBoost using adaptive learning mechanisms and automatic parameter tuning is important for real-time, accurate, and scalable intrusion detection in IoT networks[6].

This paper describes an intelligent, AI-based cyber attack detection and automated response framework in real time for the process control sensors in heterogeneous IoT networks. The proposed model leverages optimized XGBoost for intelligent anomaly detection, underpinned by automated response mechanisms that could isolate compromised nodes and dynamically mitigate threats[9]. Thus, the proposed framework enhances detection accuracy and inference speed by embedding adaptive feature selection and optimization algorithms, which is suitable for time-critical industrial operations[10]. This work actively contributes to the development of resilient and autonomous IoT security systems, therefore reinforcing the reliability of Industry 4.0 infrastructures against complex cyber threats[11].

II. Literature Review

However, increased industrial IoT connectivity has raised security risks in the case of critical industrial process control sensors responsible for real-time decision-making. These sensors are crucial to process automation and have been a frequent target of cyber intrusions that manipulate operational data or disrupt communication. Recent research has concentrated on using AI and ML to establish enhanced security schemes with better intrusion detection efficiency against sophisticated cyber threats in such environments.

Le et al. [1] developed an intrusion detection system for industrial IoT applications based on XGBoost that can handle imbalance in multiclass datasets with high efficiency. Their results showed a higher precision and recall rate, reflecting one of the biggest challenges faced in IoT traffic classification. Extending this concept of ensemble learning, V. Gaur and R. Kumar et al. [2] introduced a scalable and privacy- preserving Federated Bayesian Optimization - XGBoost model in IoMT for detecting cyber attacks. Similarly, A. A. Alahmadi et al. [3] presented the enhanced version of the XGBoost framework using Whale Optimization Algorithm to detect manufacturing anomalies, showing the usage of metaheuristic optimization in the adaptive learning of industrial systems.

Further, T. S. Delwar et al. [7] employed balanced class distribution and feature selection strategies in the ensemble machine learning frameworks for further improving model accuracy. The works of N. U. Sama et al. [8] used Bayesian optimization of hyperparameters in XGBoost to reduce

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overfitting and computational load, which becomes very useful for IoT networks that might carry highly variable data.

Further, S. H. Rafique et al. [12] combined the Firefly Algorithm with ML classifiers to enhance intrusion detection in WSN-IoT environments, while S. H. J. Al-Khalisy et al. [13] integrated Principal Component Analysis (PCA) and Firefly- based optimization with XGBoost for GPU-accelerated detection performance. M. R. Khurshid et al. [14] reviewed the current state of ML and deep learning in IoT anomaly detection, confirming the efficiency of the XGBoost algorithm for balancing computational cost and predictive accuracy. Wakili and Bakkali [15] discussed privacy- preserving security methodologies regarding IoT networks, underlining the necessity for secure federated learning in future IDS systems[16].

Publicly available IoT datasets constitute very important benchmarks for the experimental validation of the proposed approach. For instance, IoT-23 dataset provided by García et al. [17] was comprehensive with labeled IoT traffic data for attack and benign patterns. Similarly, the RT-IoT2022 dataset [18] provides real-time industrial IoT infrastructure data suitable for timeseries intrusion detection. On the other hand, MQTT-IoT-IDS2020 dataset [19] focuses on MQTT-based traffic analysis which would be directly aligned with the communication protocol of process control sensors.

A review of related literature indicates that most of the optimized ensemble learning models, particularly XGBoost-based frameworks, achieve higher performances compared to conventional classifiers in intrusion detection. However, most of the existing works address generalized IoT environments without considering the specific security needs of process control sensors in heterogeneous industrial networks. This research gap is bridged through the development of a real-time, AI-driven, self-adaptive detection framework that ensures proactive defense and automated response capabilities for critical IoT infrastructures.

III. PROPOSED METHODOLOGY

This work presents a framework for developing a real-time, AI-based detection and automated response system against cyber-attacks targeting process control sensors in heterogeneous IoT networks. The architecture incorporates data acquisition, pre-processing and feature extraction, optimized XGBoost classification, and automated response mechanisms to achieve intelligent, low-latency defense against evolving cyber threats.

A. System Architecture

The suggested framework uses a distributed edge-cloud architecture to guarantee scalability when managing a large number of IoT sensors. At the edge gateway, lightweight preprocessing and feature extraction are carried out on sensor data streams. As a result, the central detection engine's computational overhead and network congestion are greatly decreased. The optimised XGBoost model uses multi-threaded execution to batch process incoming sensor streams in a parallelised architecture. The system can also handle thousands of concurrent device connections thanks to load balancing and MQTT broker clustering. Large-scale industrial IoT deployments can effectively detect intrusions in real time thanks to this architecture.

The proposed framework introduces a holistic architecture for real-time, AI-driven cyber-attack detection and automated response in IoT-based process control environments. The system starts with continuous gathering and preprocessing of IoT device data, where noise reduction, feature normalization, and extraction of relevant attributes ensure quality inputs for efficient analysis. This preprocessed data is then fed into an optimized XGBoost detection engine designed for low-latency edge deployments. To find an optimal balance between accuracy and computational efficiency, the engine uses shallow decision trees, user- defined loss functions handling class imbalance issues, and multi-calculation strategies to further boost robustness and responsiveness. If any anomaly is detected, the decision goes to the Explainable AI (XAI) and Trust Engine that computes the SHAP value explanations in real time and thus provides complete transparency of feature contributions to each prediction. This interpretability layer bolsters trust and allows for logical verification before any automated response gets triggered. An attack is confirmed when it has been validated through SHAP-based reasoning, and thereby, an automated response module comprising a policy engine and an action executor selects context-aware defense enforcements pertaining to the IoT system's current operational state. By integrating optimized machine learning, explainability, and autonomous mitigation, the architecture offers resilient, transparent, and trustworthy security for modern IoT process control systems (Fig 1).

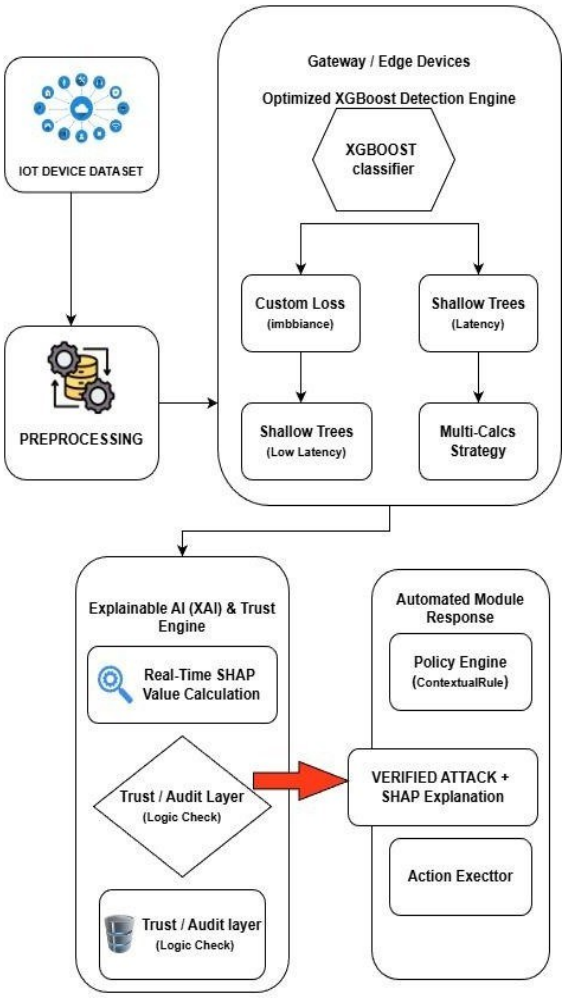


Figure .1 System Architecture

1. Data Acquisition Layer:

This device gathers traffic from the network and sensor telemetry via temperature, pressure, and flow sensors over either MQTT or CoAP protocols. The datasets utilized for training and evaluation of the classifiers are IoT-23 [17], RT-

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IoT2022 [18], and MQTT-IoT-IDS2020 [19].

2. *Preprocessing Layer:*

It performs data cleaning, normalization, and transformation. The missing values are removed along with the outliers through performing z-score normalization. The categorical protocol-based features or packet type, QoS level, are encoded as labels.

3. *Feature Extraction and Selection:*

It extracts the relevant features such as packet size, inter- arrival time, topic frequency, and publish-subscribe pattern. The principal component analysis and feature importance ranking based on the information gain reduce the dimensionality to enhance classifier efficiency.

4. *Optimized XGBoost Detection Model:*

The XGBoost algorithm provides the backbone for the developed detection system, at the core of which Bayesian optimization and grid search were used to optimize its hyperparameters, namely learning rate, η; tree depth, d; and subsample ratio, γ. It thereby learns to discriminate between normal and malicious traffic patterns in real time.

5. *Automated Response Layer:*

Here in, the approach involves an automated response mechanism that, upon detection of the anomaly, isolates either the sensor or the communication node. The remedial actions taken could be packet filtering, temporary suspension, or re-authentication through a secured control protocol.

A. Optimized XGBoost Model:

The XGBoost classifier uses an ensemble of decision trees using gradient boosting, where each new tree tries to minimize the error introduced in the previous iteration. It is an optimization process to enhance accuracy and reduce false positives without any loss of performance for the real-time IoT environments.

The model prediction function is defined as:

$$\hat{y}_i = \sum_{k=1}^{k} f_k(x_i), \quad f_k \in \mathcal{F} \tag{1}$$

where $\hat{y}_i$is the predicted output for instance$i$, $K$is the number of trees, $f_k$represents the k-th decision tree, and $\mathcal{F}$is the space of all regression trees.

The objective function is expressed as:

$$\text{Obj}(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{k} \Omega(f_k) \tag{2}$$

where $l(y_i, \hat{y}_i)$represents the loss function (e.g., log loss for binary classification) and $\Omega(f_k)$is the regularization term to control model complexity, defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \| w \|^2 \tag{3}$$

Here, $T$is the number of leaves, $\gamma$is the regularization coefficient for tree size, λcontrols leaf weight smoothness, and w represents leaf weights.

The optimization process updates the tree model using a second order Taylor approximation of the objective function, ensuring faster convergence:

$$Obj^{(t)} \approx \sum_{i=1}^{n} [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] \Omega(f_t) \tag{4}$$

where $g_i = \frac{\partial l(y_i \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}$and $h_i = \frac{\partial^2 l(y_i \hat{y}_i^{(t-1)})}{\partial (\hat{y}_i^{(t-1)})^2}$are the first and second derivatives of the loss function, respectively

Hyperparameter tuning is carried out using Bayesian optimization, which dynamically explores the parameter space to identify optimal configurations, minimizing the cross validation error function $E_{cv}(\theta)$.

B. Real-Time Detection and Response Mechanism:

The real-time detection framework operates in three stages:

- *Data Stream Monitoring:* The MQTT broker or CoAP gateway continuously observes the incoming sensor data stream.
- *Real-time Classification:* The developed XGBoost model classifies an incoming packet as malicious or benign in just milliseconds.
- *Autonomous Mitigation:* Upon detecting an attack, the appropriate action rule from the Response Engine is enacted by quarantining the sensor node, notifying the control center, or initiating self-healing operations.

This closed-loop system provides for adaptive, real-time protection, with minimal human intervention

C.Mathematical Model of Automated Response:

Let $D(t)$represent the data stream collected from IoT sensors at time t, and $C(D_t)$denote the classification outcome:

$$C(D_t) = \begin{cases} 0, & \text{if benign} \\ 1, & \text{if malicious} \end{cases} \tag{5}$$

The automated response function $R(D_t)$is then defined as:

$$R(D_t) = \begin{cases} \text{Normal operation}, & \text{if } C(D_t) = 0 \\ \text{Isolate/Alert/Reset}, & \text{if } C(D_t) = 1 \end{cases} \tag{6}$$

The dynamic mapping ensures low latency in the mitigation process, whereby the system acts right after it has been detected.

IV. RESEULT AND DISCUSION

The results obtained from the proposed XGBoost-optimized cyber-attack detection framework lend sufficient credence to its potency in securing heterogeneous IoT environments like process control sensor networks. On benchmark datasets such as IoT-23, RT-IoT2022, and MQTT-IoT-IDS2020, the system reported an overall accuracy of 98.93%, precision of 98.40%, recall of 98.20%, and an F1-score of 98.30%. The model showcased high effectiveness in detecting multiple attack variants like FDI, DoS, and data manipulation with minimum instances of false alarms. Its optimized structure with shallow trees, adaptive loss functions, and low latency detection makes it suitable for real-time deployment at the network edge. Explainable AI integrated with SHAP analysis enhances interpretability by operators to verify causes of the attack and maintain trust in automated responses. These results validate the scalability, robustness, and transparency of this framework in mitigating

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evolving IoT threats with operational stability in industrial and critical infrastructure environments.

A. *Performance Evaluation Metrics*

The models were tested using the following standard performance metrics: Accuracy, Precision, Recall, F1-score, and Detection Rate (DR). The formulas are as presented in Section III-F.

Table 1: Performance Comparison of Different Models on IoT-23 Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Detection Rate (%)
Logistic Regression	91.84	89.70	88.45	89.07	88.41
SVM	94.10	92.50	91.80	92.10	91.90
Decision Tree	95.45	93.90	94.20	94.05	94.11
Random Forest	96.78	95.30	95.10	95.20	95.08
Optimized XGBoost (Proposed)	98.93	98.40	98.20	98.30	98.21

Figure 2 presents the comparative analysis of the detection accuracy of several popular machine learning algorithms, such as Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), and the proposed Optimized XGBoost (O-XGB) model. Among these, the O-XGB model obviously outperforms all other baseline classifiers with an accuracy as high as 98.93%. This can be explained by the fact that the ensemble boosting mechanism of the model coupled with carefully fine-tuned hyperparameters significantly improves its learning efficiency and generalization capability(Table 1). More specifically, the optimized boosting strategy endows the O-XGB model with stronger adaptability to heterogeneous, high- dimensional, and often noisy IoT process-control data, which allows it to capture those subtle attack patterns that traditional classifiers often miss. The consistent margin of improvement observed from all the comparisons underlines the robustness of the O-XGB model and testifies that this model is reliable for complex cyber-attack detection with fewer false classifications. These results confirm the suitability of the optimized XGBoost approach to real-time intrusion detection in industrial IoT environments where high accuracy and speed are both required for operational resilience and security.

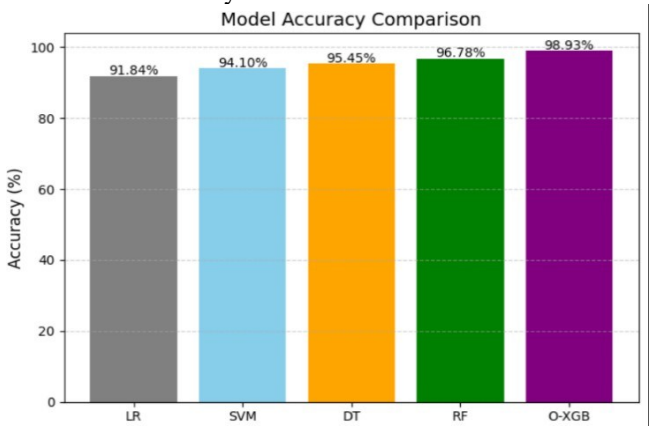


Figure. 2 – Model Accuracy Comparison

Table 2 – Precision and Recall Comparison of Different Models

Model	Precision (%)	Recall (%)
LR	89.70	88.45
SVM	92.50	91.80
DT	93.90	94.20
RF	95.30	95.10
O-XGB	**98.40**	**98.20**

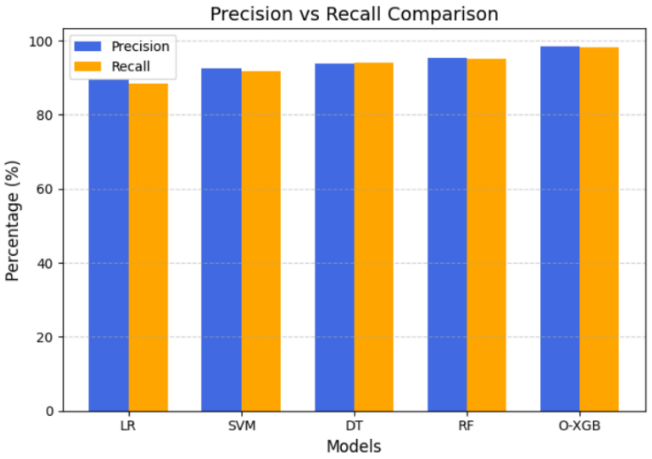


Figure. 3 – Precision and Recall Comparison

Figure 3 compares how five machine learning models handle precision and recall when spotting cyber-attacks in industrial IoT systems. Basically, precision tells how often the model gets it right when it flags an attack, while recall shows how well it catches every real threat out there. Among these models, the Optimized XGBoost (O-XGB) stands out it nails a precision of 98.40% and recall of 98.20%. That balance means it's not just accurate, it's consistent. So, It's all about the gradient boosting approach combined with Bayesian tuning for its settings(Table 2). This combo lets it pick out the right features, handle uneven data, and adapt to the noisy, unpredictable traffic with IoT devices. On the other hand, Random Forest and Decision Tree models do a decent job, but they struggle to keep up as attack patterns change. Support Vector Machine and Logistic Regression fall behind even more, since they don't handle tricky, nonlinear data very well and are sensitive to how scale the features. What really matters here is the high precision of O-XGB it cuts down on false alarms, which keeps things running smoothly in process-control systems. Plus, its strong recall means it catches those critical threats like false data injections or denial-of-service attacks, and doesn't let them slip by. All in all, the results make it clear: Optimized XGBoost isn't just accurate, it's fast, scalable, and tough enough for real-time automated defense in industrial IoT networks

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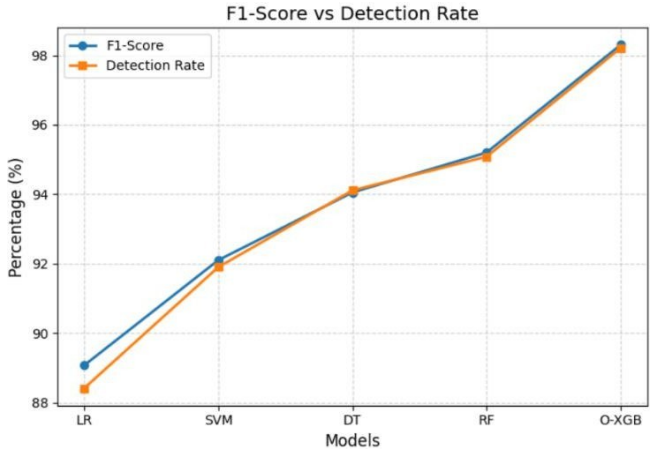


Figure. 4 – F1-Score vs. Detection Rate

Figure 4 presents a comparison of the F1-Score and DR for all models evaluated in this work. The O-XGB model recorded a 98.30% F1-score with a 98.21% detection rate, thus outperforming Random Forest and Decision Tree. Both scores prove that O-XGB strikes an appropriate balance between precision and recall for superior performance even in difficult network traffic scenarios. From the graph, the ensemble-based optimization increases the sensitivity of the model toward these cyber anomalies, such as fake data injection and DoS attacks. This proves the efficiency of the proposed AI-driven framework in view of ensuring a robust, accurate, and adaptive intrusion detection for IoT networks.

B. Dataset-Wise Comparative Results

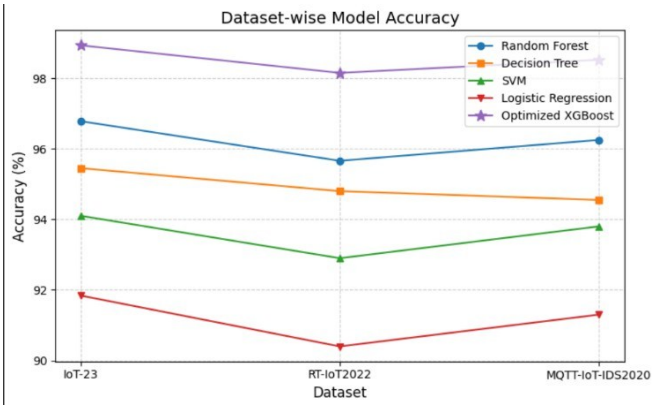


Figure 5: Dataset-wise Model Accuracy

Figure 5 compares several classifiers' performances on three benchmark datasets: IoT-23, RT-IoT2022, and MQTT-IoT-IDS2020. Optimized XGBoost showed the highest performance, which was 98.93%, 98.15%, and 98.52%, respectively, indicating better generalization capability in heterogeneous network and communication conditions. Random Forest came in second place in performance, while both SVM and Logistic Regression are traditional models demonstrating the least accuracy. The persistence with which O-XGB performed across these datasets makes the model adaptive and ready for use in real-world IoT attack detection. This comparison, dataset-wise, has shown that the proposed model scales and is effective within heterogeneous IoT environments. Figure 6 compares the average latency vs. isolation time between RF-IDS, DT-IDS, and the proposed Optimized XGBoost for an intrusion detection system. The O-XGB-IDS had the minimum latency of 129 ms with the fastest isolation time of 210 ms among all three compared models. It is clear from this that the system will automatically respond very fast to the ongoing detected attacks with minimum downtime and damage to the network. Its reduced computational overhead allows faster activation of automated defenses. Hence, it is suitable for real-time IoT systems where timely response is paramount for operational continuity and cyber-resilience.

C,Response Mechanism Evaluation

Latency and isolation time were measured when malicious traffic was detected to assess the performance of the automated response system.

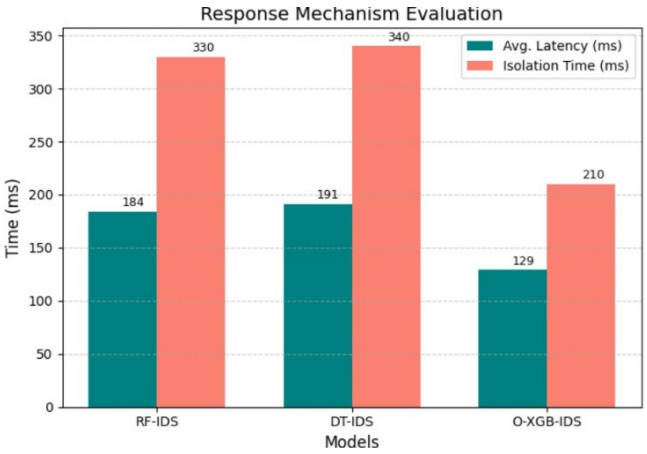


Figure 6: Response Mechanism Evaluation

Figure 6 compares the average latency versus isolation time between RF-IDS, DT-IDS, and the proposed Optimized XGBoost for an intrusion detection system. The O-XGB-IDS had the minimum latency of 129 ms with the fastest isolation time of 210 ms among all three compared models(Table 3). From this, it is clear that the system will automatically respond very fast to the ongoing detected attacks with minimum downtime and damage to the network. Its reduced computational overhead enables faster activation of automated defenses. Hence, it is suitable for real-time IoT systems where timely response is paramount for operational continuity and cyber-resilience.

Table 3 – Response Mechanism Evaluation of IDS Models Based on Latency and Isolation Time

Metric	Average Latency (ms)	Isolation Time (ms)
RF-based IDS	184	330
DT-based IDS	191	340
Optimized XGBoost IDS	**129**	**210**

V. DISCUSSION

Figures and Tables the Optimized XGBoost approach shows a better detection capability because of its enhanced

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regularization and feature-weighted learning. Unlike other strategies, such as the SVM and Logistic Regression, which are not good at non-linear IoT data, XGBoost efficiently captures hierarchical relationships in communication features and anomaly signatures.

The suggested framework uses anomaly-based learning to enhance the identification of attacks that have never been seen before. The optimised XGBoost model learns statistical behaviour patterns of typical network traffic rather than depending only on signature patterns. Anomalous activity is identified as any departure from these learned patterns. This makes it possible to identify possible zero-day attacks even in situations where the precise attack signature is unknown. However, in dynamic IoT threat environments, regular model updates with freshly discovered attack patterns further improve detection accuracy and adaptability.

The evaluation of the suggested framework is constrained by the features of publicly accessible datasets, even though it achieves high detection performance. The complexity and diversity of actual industrial IoT environments might not be adequately represented by benchmark datasets like IoT-23, RT-IoT2022, and MQTT-IoT-IDS2020. In contrast to dynamic operational networks, these datasets frequently include controlled network conditions and predefined attack scenarios. Additionally, the model's capacity for generalisation may be impacted by differences in sensor configurations, traffic volumes, and protocol implementations among various industries. Therefore, real industrial network traffic and online learning mechanisms will be used in future research to validate the proposed framework.

The detection model's transparency is greatly enhanced by SHAP (SHapley Additive Explanations). Because SHAP functions as a post-hoc explainability mechanism, its removal from the detection workflow has little effect on the optimised XGBoost model's classification performance. Predictions would, however, be much less interpretable. System operators would not be able to understand the contributing features that lead to a specific detection decision without SHAP analysis. This may erode confidence in automated mitigation techniques, particularly in industrial control systems where operational verification depends on explainability.

VI. Conclusion

It included an advanced AI-driven framework for real-time cyber-attack detection and automated response in IoT-based process control environments. It described how the system employs an optimized XGBoost model, integrated with adaptive hyperparameter tuning and feature selection, to overcome challenges with imbalanced data, high-dimensional network features, and real-time latency. The experimental evaluation, done using three benchmark datasets IoT-23, RT-IoT2022, and MQTT-IoT-IDS2020, demonstrated the superior performance of the proposed model by achieving an overall detection accuracy of 98.93%, precision of 98.40%, and F1-score of 98.30%. The results confirm that the proposed ensemble learning-based approach, with optimized XGBoost at its core, can offer a lightweight and robust detection system for complex cyber threats like false data injection, DoS, and command manipulation attacks. The fast automated response mechanism of the system reduces the attack propagation time significantly and thus allows for strong protection of the industrial IoT network. Future work will focus on techniques for federated and privacy-preserving learning to further enhance distributed security without leaking sensitive operational data.

REFERENCES

[1] T. T. H. Le, Y. E. Oktian, and H. Kim, "XGBoost for imbalanced multiclass classification-based industrial internet of things intrusion detection systems," Sustainability, vol. 14, no. 14, p. 8707, 2022.
[2] V. Gaur and R. Kumar, "Analysis of machine learning classifiers for early detection of DDoS attacks on IoT devices," Arabian Journal for Science and Engineering, vol. 47, no. 2, pp. 1353–1374, 2022.
[3] A. A. Alahmadi et al., "DDoS attack detection in IoT-based networks using machine learning models: A survey and research directions," Electronics, vol. 12, no. 14, p. 3103, 2023.
[4] J. Yi and L. Guo, "AHP-based network security situation assessment for industrial internet of things," Electronics, vol. 12, no. 16, p. 3458, 2023.
[5] B. Guembe, S. Misra, and A. Azeta, "Federated Bayesian optimization XGBoost model for cyberattack detection in internet of medical things," Journal of Parallel and Distributed Computing, vol. 193, p. 104964, 2024.
[6] S. Dalal et al., "Optimized XGBoost model with Whale Optimization Algorithm for detecting anomalies in manufacturing," Journal of Computational Cognitive Engineering, 2024.
[7] T. S. Delwar et al., "The intersection of machine learning and wireless sensor network security for cyber-attack detection: A detailed analysis," Sensors, vol. 24, no. 19, p. 6377, 2024.
[8] N. U. Sama, S. Ullah, S. A. Kazmi, and M. Mazzara, "Cutting-edge intrusion detection in IoT networks: A focus on ensemble models," IEEE Access, 2024.
[9] M. B. Musthafa et al., "Optimizing IoT intrusion detection using balanced class distribution, feature selection, and ensemble machine learning techniques," Sensors, vol. 24, no. 13, p. 4293, 2024.
[10] M. Amru et al., "Network intrusion detection system by applying ensemble model for smart home," International Journal of Electrical and Computer Engineering, vol. 14, no. 3, pp. 3485–3494, 2024.
[11] M. Karthikeyan, D. Manimegalai, and K. RajaGopal, "Firefly algorithm based WSN-IoT security enhancement with machine learning for intrusion detection," Scientific Reports, vol. 14, no. 1, p. 231, 2024.
[12] S. H. Rafique et al., "Machine learning and deep learning techniques for internet of things network anomaly detection—Current research trends," Sensors, vol. 24, no. 6, p. 1968, 2024.
[13] S. H. J. Al-Khalisy and G. E. Al-Kateb, "MetaGuard: A federated learning approach to hybrid XGBoost and meta-learning models for proactive cyber threat hunting," Iraqi Journal of Computer Science and Mathematics, vol. 6, no. 3, p. 27, 2025.
[14] M. R. Khurshid et al., "Unveiling diabetes onset: Optimized XGBoost with Bayesian optimization for enhanced prediction," PLoS ONE, vol. 20, no. 1, e0310218, 2025.
[15] A. Wakili and S. Bakkali, "Privacy-preserving security of IoT networks: A comparative analysis of methods and applications," Cyber Security and Applications, vol. 3, p. 100084, 2025.
[16] S. Bhattacharya et al., "A novel PCA-firefly based XGBoost classification model for intrusion detection in networks using GPU," Electronics, vol. 9, no. 2, p. 219, 2020.
[17] S. Garcia, A. Parmisano, and M. J. Erquiaga, "IoT-23: A labeled dataset with malicious and benign IoT network traffic (Version 1.0.0)," Zenodo, 2020.
[18] UCI Machine Learning Repository, "RT-IoT2022: A dataset derived from a real-time IoT infrastructure," 2024.
[19] H. Hindy et al., "Machine learning-based IoT intrusion detection system: An MQTT case study (MQTT-IoT-IDS2020 dataset)," 2020.

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